

Utrecht Summer School
Introduction to Complex Systems

Wednesday projects

Network dynamics: Epidemics & blackouts

1 Infectious disease spreading (*Python*)

In this project, we will study a simple model of spreading on a network, which will illustrate how for example infectious diseases or ideas can spread through a population. The focus in this project will be on the effect of the network structure on the spreading through the network.

In this project's simplified model, each node of the network – representing e.g. a person in a population – can either be infected (1) or uninfected (0). When a node has become infected, it will stay infected. We initialize the network by selecting randomly one of the nodes to be infected. Then, at every timestep, we look at all the uninfected nodes that are neighbours of an infected node. Each of these uninfected nodes then becomes infected with probability p .

Use the files `infection.ipynb` and `infection_functions.py` and put them in the same folder. Again, only the main notebook needs to be opened and amended in this project.

NOTE: The files contain a function called `NetworkProperties` that can be called upon to visualise the network structure and to compute the degree of the nodes in the network when it is given an adjacency matrix A . It can be called by using `deg_av, degs=NetworkProperties(A)`. The variable `deg_av` gives the average degree of the nodes in the network and `degs` contains a list of the degree of the individual nodes.

1.1 Spreading on a lattice network

First, set-up the parameters and settings such that spread on a 16×16 lattice network is modelled. The network type can be changed by changing the variable `network_type`. Now, if you run the code, you will see a figure with the network structure in which all infected nodes will be marked in red, and in another panel the fraction of the infected nodes in the network. The programmed lattice network has periodic boundaries. Can you explain the visualisation of the network?

Run the code a couple of times. Why does the exact results change each time? Vary the infection probability p . How does this change the results? Does it take longer to fully infect the whole network. Try to quantify the behaviour for different values of p . To prevent randomness of a single run, it is better to take the average over multiple runs per p -value.

NOTE: you can speed up the simulations by taking a higher value for `plot_interval`, so plots are refreshed less often. You can also completely shut off the visualisations by setting `showPlot` and/or `showGraph` to false, which further speeds up the simulation.

1.2 Spreading on a random network

Instead of the very regular lattice network, now consider a random network that is created as follows. We start with a regular lattice network. Then, with probability q a link of the lattice network is broken, and one of the connecting nodes of that link makes a new link with a randomly chosen node. How does this procedure change the network structure? Does the degree of nodes change? Does the average degree change? Run the infection model on such random networks (you only have to change the options for this). Investigate the effect of different q and p combinations. Compare the results on this random network to the results on the lattice network. Can you physically explain the differences between the two?

1.3 Spreading on a community network

Now, consider a community network that consists of two communities: that is, two random networks (created as before) with r links between these communities. (In the code, these inter-network links are created while simultaneously breaking the same number of intra-network links.) Determine how this procedure changes the network structure? Does the degree of nodes change? Does the average degree change?

Run the model for a community network of two communities of size 128 each (to have the same total number of nodes). Study what happens when you change r , q and p . Compare your results with your previous results on the lattice network and the random network.

1.4 Additional suggestions

If time permits, feel free to experiment with different network structures. For instance, you could try to increase the number of communities. Alternatively, you can construct a custom adjacency matrix to use in the simulations.

In case you have even more time, you could try to modify the source code in `infection_functions.py` to change the node dynamics in a way that infected people can recover after a while and become uninfected again.

NOTE: If you are using your own adjacency matrix, be aware that the visualisation only works if the adjacency matrix is symmetric. That is, the visualisation only works properly for undirected networks. However, the internal code for the simulation should work even for directed networks.

2 Power grid (*Python*)

In this part we will study a conceptual model of a power grid. The nodes of the network are consumers or producers, and the edges of the network represent the power lines between connected nodes. At each node, the voltage and current oscillate at a frequency of around 50 Hz (50 times per second), known as alternating current (AC). Stable operation of the power grid requires that all nodes are synchronized to this frequency (± 0.02 Hz or so). If the frequency becomes too high or low at any node, this can cause a blackout since power flows can grow too large.

Due to the alternating current, we can idealize each consumer/producer i as an oscillator with a natural frequency $\Omega = 50$ Hz. The natural frequency is the frequency at which the oscillator would swing back and forth when isolated from external influences. The state of each oscillator is defined by its phase angle $\phi_i(t)$. For power grids, it is convenient to look at the phase angle deviation from 50 Hz, which we denote $\theta_i(t) = \phi_i(t) - \Omega$. In this coordinate system, frequency synchronisation of the grid corresponds to a fixed point, $\dot{\theta}_i(t) = 0$ for all i .

Now suppose that each node has a power input or output P_i , representing production ($P_i > 0$) or consumption ($P_i < 0$). For a stable grid, power flows must be balanced, i.e. all P_i must sum to zero. The power flow between two nodes i and j depends on the sine of their phase difference, $\sin(\theta_j - \theta_i)$.

Such a system can be mathematically described by the Kuramoto model, governed by

$$\frac{d\theta_i}{dt} = P_i + \frac{K}{N} \sum_j A_{ij} \sin(\theta_j - \theta_i). \quad (1)$$

Here K is proportional to the overall line capacity (effectively acting as the coupling strength) and A_{ij} are the coefficients of an adjacency matrix A that indicate which oscillators are connected by a power line.

Use the files `power_grid.ipynb` and `power_grid.py` (put them in the same folder). You only need to edit the Jupyter notebook to do this exercise.

2.1 Fully coupled network

The Kuramoto model corresponds to the situation where each oscillator is coupled to each other oscillator, that is $a_{ij} = 1$ if $i \neq j$. Thus, the standard adjacency matrix for the Kuramoto model is

$$A = \text{np.ones}(N, N) - \text{np.eye}(N).$$

Using this network structure (fully coupled network), run the code (`simulate` function) for various coupling strengths and describe the degree of synchronization. You can look at the evolution of the phases θ_i and the order parameter $r(t)$ that are plotted at the end of the simulation to get a feel of the amount of synchronisation. Feel free to also play around with the amount of oscillators N . Beware though that simulations get slower as N increases, and takes up more computer memory.

2.2 Order parameter

The (instantaneous) order parameter $r(t)$ has already been implemented in the code as follows

$$r(t) = \frac{1}{N} \left| \sum_{j=1}^N e^{i\theta_j(t)} \right|, \quad (2)$$

where i is here the imaginary unit (look up complex numbers if you don't know what that is). Does this value help you to accurately quantify the amount of synchronization? Why or why not?

It is possible to define a final order parameter $\bar{r}(t)$ by taking the long-time average over the instantaneous order parameter $r(t)$. Specifically, $\bar{r}(t)$ can be defined as

$$\bar{r} = \frac{1}{T} \lim_{T \rightarrow \infty} \int_{T_0}^{T_0+T} r(t) dt, \quad (3)$$

where T_0 is spin-up time period. That is, you ignore the value of the instantaneous order parameter $r(t)$ for times $t < T_0$. Why would we want something like this?

Now, implement the computation of $\bar{r}$ at the end of the script. To do so, you can just take the average of the order parameter $r(t)$ over the last part of your simulation (use the function `mean` or `np.mean` for this). Then, compute $\bar{r}$ for various coupling strengths and plot them as function of K . Do you see a phase transition? What does this imply for the design of a power grid?

Tip: if you set `live_plot` to `False`, the code will run faster, as the visualisation steps are skipped.

2.3 Effect of the network structure on synchronization

So far, we have used a fully connected network where $a_{ij} = 1$ if $i \neq j$. This is not very realistic for a power grid. Investigate the effect of other network structures on power grid stability.

Two suggestions are:

- Construct Watts-Strogatz networks (an example is in the notebook) and investigate synchronisation as a function of the parameters p (rewiring probability), k (average degree) and N (system size).
- Construct your own network topology for a small power grid (N between 5-10) and distribute the power inputs/outputs P_i such that they sum to zero. Often, consumers are on dead ends of the network while producers have a higher degree, acting as hubs. Keep the P_i fixed but vary the network structure. You can also make the adjacency matrix weighted to represent power lines of different capacity. Can one network be synchronized for lower K than another? What would that mean for the cost of building the power grid?

2.3.1 Effect of a line failure

Start with a network structure A and power distribution P_i that leads to a stable synchronized state. Then, remove one line (try different ones). Do the oscillators still synchronize? If not, what have you observed? What does this mean for the context of a power grid?

3 Cascading failure (*Julia*)

Follow the instructions in the notebook `cascade_blackout.ipynb`. A plain Julia script version is available in `cascading_failure.jl` (based on this example.)